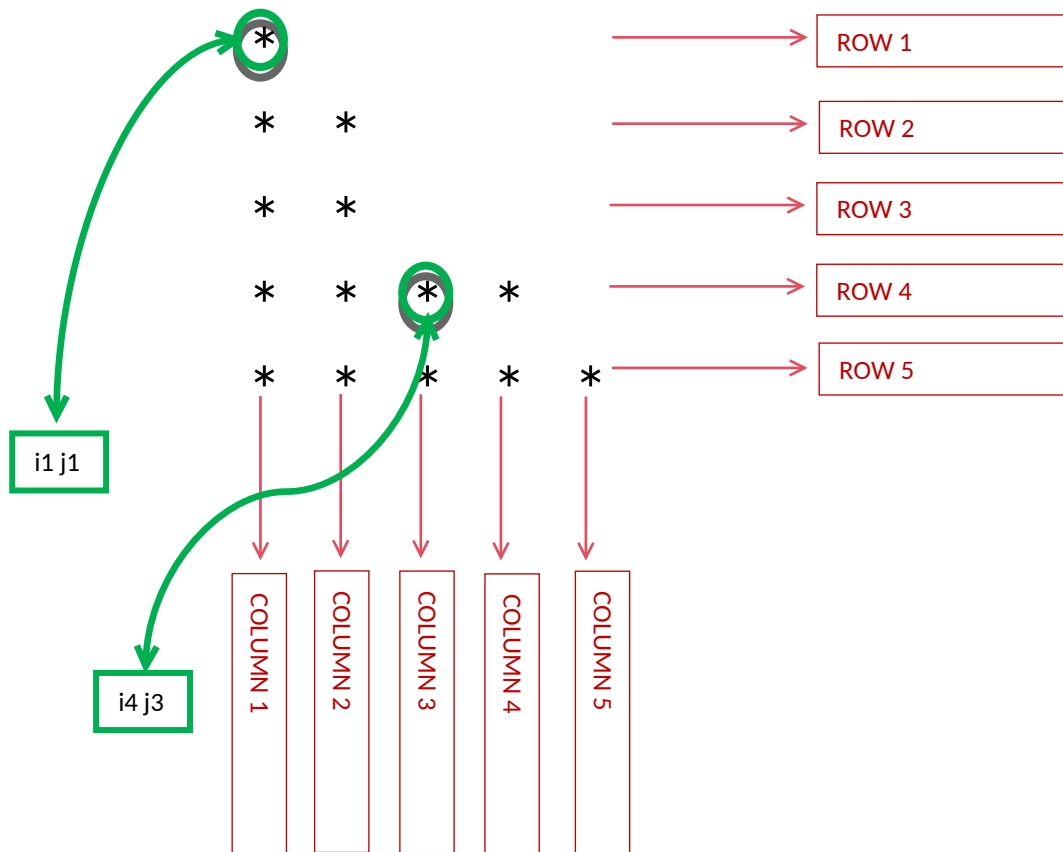


WEEK 01 MODULE
Programming Practice

TOPIC : 1 'Star Pattern'

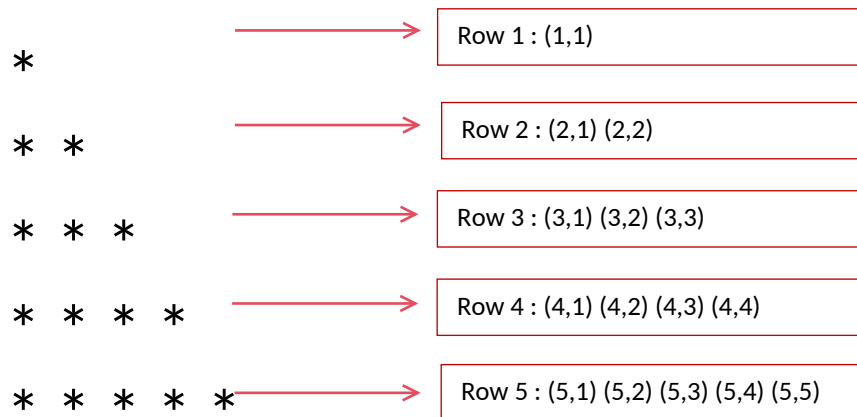
4. HALF DIAMOND(UPPER) $\longleftrightarrow$ RIGHT - ANGLED TRIANGLE



1. Problem Analysis :-

i. Pattern Structure:

- The output forms an upper half diamond
- Unlike the square patterns, the number of stars is not constant across all rows.
- The pattern starts with one star in the first row and increases by one star in each subsequent row until the last row contains n stars.
- For n=5



ii. Execution Flow:

Step 1: Observe the Grid Concept:

- The pattern can be visualized as a triangular portion of an $n \times n$ grid. Each row prints only a subset of the available columns.

- size = $n \times n$.

- Each cell in the grid represents a unique coordinate (i, j), where i denotes the row index and j denotes the column index.

- For $n = 5$:

The number of stars in each row depends directly on the row number.

Observation:

- Row 1 → 1 star
-
- Row 2 → 2 stars
-
- Row 3 → 3 stars
-
- Row 4 → 4 stars
-
- Row 5 → 5 stars
-

A clear relationship can be observed:

Number of stars = Row Number

Therefore:

- Row 1 prints 1 star
- Row 2 prints 2 stars
- Row 3 prints 3 stars
-
- Row n prints n stars
-

Step 2: Traverse the grid row by row using outer loop.

- The outer loop is responsible for generating rows.

Step 3: For every row, determine how many stars need to be printed.
For every row, determine how many stars need to be printed.

- The number of stars required is equal to the current row number.

Step 4: Use the inner loop to print stars from column 1 up to the current row number.

Step 5: After all stars of the current row are printed, move to the next line.

Step 6: Repeat the same process until all n rows have been completed.

- The pattern growth is controlled by the inner loop limit.

- the inner loop runs from 1 to the current row number:

$j \leq i$ (This automatically increases the number of stars printed in each row.)

{

for($j = 1; j \leq i; j++$)

Meaning:

Current row gets as many stars as its row number.

}

Half Diamond Upper follows the same logic as the Triangle Pattern. The number of stars printed in each row is equal to the row number.

2. Program :-

```
package week1_module;
import java.util.Scanner;

public class StarHalfDiamondUpper {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the number of rows: ");
        int n = sc.nextInt();

        for(int i = 1; i <= n; i++) {

            for(int j = 1; j <= i; j++) {
                System.out.print("* ");
            }

            System.out.println();
        }

        sc.close();
    }
}
```

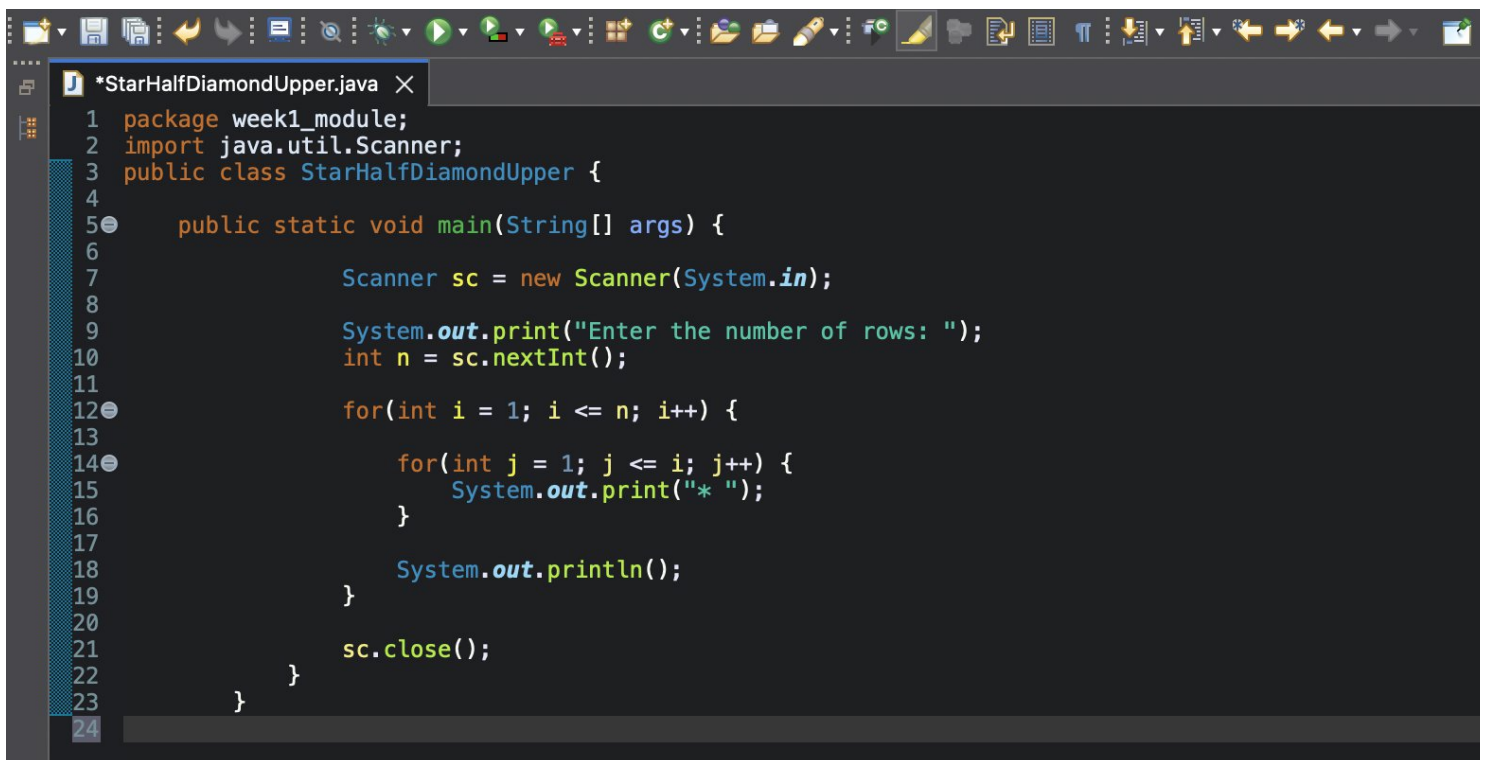
3. Program Output :-

Enter the size of the rows: 5

```
*
* *
* * *
* * * *
* * * * *
```

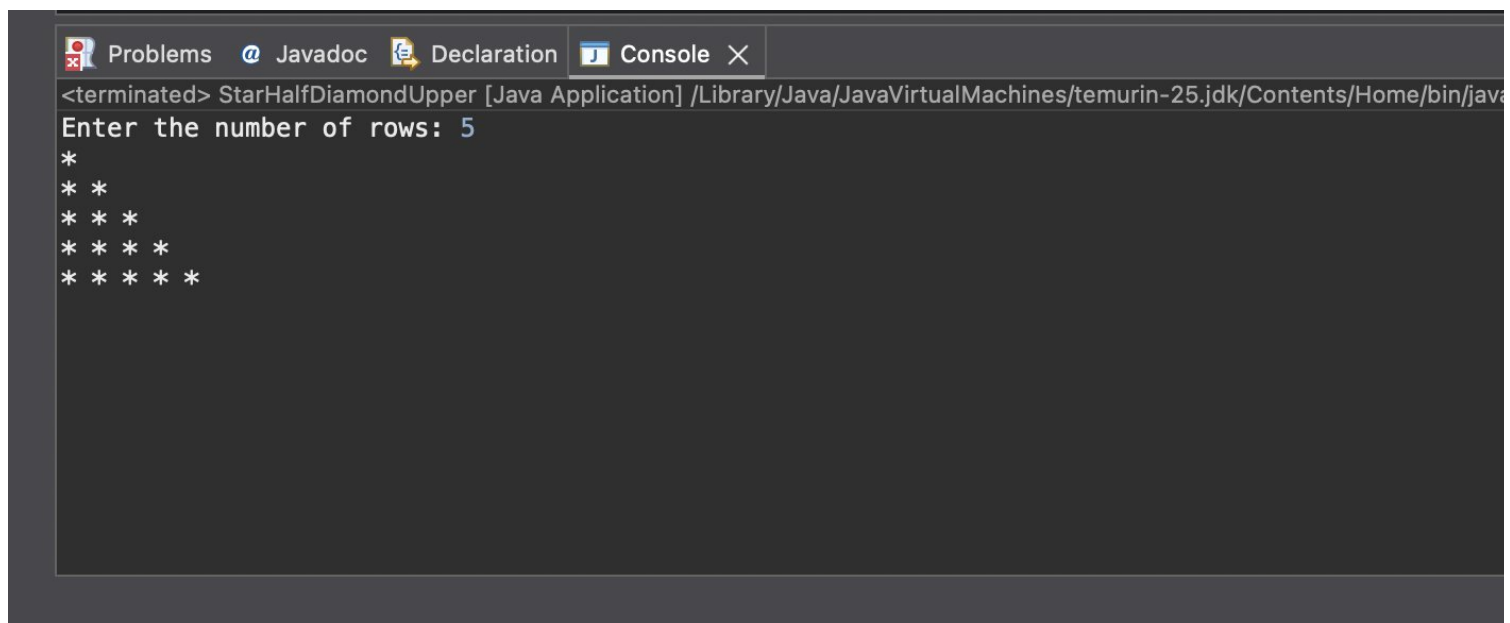
4. ScreenShot :-

i. Program :-



```
1 package week1_module;
2 import java.util.Scanner;
3 public class StarHalfDiamondUpper {
4
5     public static void main(String[] args) {
6
7         Scanner sc = new Scanner(System.in);
8
9         System.out.print("Enter the number of rows: ");
10        int n = sc.nextInt();
11
12        for(int i = 1; i <= n; i++) {
13
14            for(int j = 1; j <= i; j++) {
15                System.out.print("* ");
16            }
17
18            System.out.println();
19        }
20
21        sc.close();
22    }
23 }
24
```

ii. Program's Output :-



```
<terminated> StarHalfDiamondUpper [Java Application] /Library/Java/JavaVirtualMachines/temurin-25.jdk/Contents/Home/bin/java
Enter the number of rows: 5
*
* *
* * *
* * * *
* * * * *
```